



SUPERIOR UNIVERSITY

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Section	4A
Semester	4 TH
Subject	Programing for ai (Lab)
DOCUMENTATION OF PROJECT OF PROGRAMING FOR AI	
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DOCUMENTATION OF PROJECT OF AI

Project Title Real-Time Facial Emotion Detection Web App.

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Objective Detect and display the user's facial emotion in real time using a webcam.

Technologies Used

- Python
- Flask (web framework)
- OpenCV (camera & face detection)
- DeepFace (emotion recognition)
- HTML + CSS (frontend)

Features

- Live webcam feed in browser
- Detects 7 emotions: Happy, Sad, Angry, Fear, Surprise, Disgust, Neutral
- Draws orange box around face
- Shows dominant emotion in green text

- Fully local – no data leaves your computer
- Beautiful gradient UI

Project Files

1. app.py – Main Flask + OpenCV + DeepFace code
2. templates/index.html – Web page
3. static/style.css – Styling

How to Run

Bash

```
pip install flask opencv-python deepface  
python app.py
```

Open <http://127.0.0.1:5000> → Allow camera → Done!

Output Live video with face box + real-time emotion label.

Made with DeepFace + Flask + OpenCV

